RESEARCH LANDSCAPE REPORT — COMPUTATIONAL COUPLING

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Status: planning document, superseded. This was written at the start of the project (September 2026), before any experiment was run, and it records the research plan and novelty analysis as they stood then. It is **not** a statement of findings. Several of its premises did not survive: the human H01 microscale tier was withdrawn when its connectivity graph turned out to be empty, the CoupledEventNet / neuromorphic-efficiency direction was withdrawn along with the fabricated silicon figures supporting it, and the central motif hypothesis it sets out was tested and came out with the opposite sign.

For what the project actually found, see `../info.md` and the manuscript. For what was withdrawn and why, see `../AUDIT.md`. This file is kept because the retraction is only checkable against the plan it replaced.

Cross-Species & Cross-Scale Connectome Analysis, Circuit Remodeling, and Event-Driven Computing

Project: COMPUTATIONAL COUPLING
Lead Autonomous Research Engineer: Antigravity
Milestone: Milestone 0 Deliverable (Extended with Human Neural Data)
Date: September 17, 2026
Hardware Profile: 13th Gen Intel Core i7-13620H (16 vCPUs), 16 GB RAM (7.9 GB available), NVIDIA GeForce RTX 4050 Laptop GPU (6 GB VRAM, CUDA 12.4), 268 GB Storage.

Executive Summary

This report establishes the empirical, algorithmic, and theoretical foundations for **COMPUTATIONAL COUPLING**, extending the original three-organism scope to incorporate human neural data at two distinct biological scales: the **H01 human cortical micro-connectome** (Lichtman Lab / Harvard / Google,

Science 2024) and the **Human Connectome Project (HCP)** whole-brain macro-connectome (Washington University / NIH, *NeuroImage* 2012; *Nature* 2016).

Our comprehensive literature and prior-art audit across more than 50 seminal works in systems neuroscience, connectomics, and neuromorphic engineering yields five foundational conclusions:

1. **The Biological Scale Ladder:** Biological neural connectivity exists across fundamentally distinct organizational tiers. We establish a strict hierarchical ladder spanning *C. elegans* (~302 neurons, whole-animal), *Drosophila melanogaster* (~139k neurons, whole-brain), *Mus musculus* (~1,000 proofread cortical neurons, connectivity + in vivo calcium activity + visual stimulus), *Homo sapiens* H01 (~16,000 neurons, ~150M synapses in 1 mm^3 temporal cortex), and *Homo sapiens* HCP (180–1015 cortical parcels, macro-scale tractography and fMRI).
2. **Absolute Novelty & Forbidden Over-Claims:** We enforce strict scientific integrity. We explicitly forbid claiming that "the complete human brain connectome is available" (only regional macro-connectivity and a 1 mm^3 micro-fragment exist), that "we invented cross-species motif counting" (Lin et al., *Nature* 2024 did this descriptively), or that "we built the first whole-brain spiking model" (Shiu et al., *Nature* 2024 did this for the fly).
3. **The Core Defensible Gap:** Existing research treats connectomes either as static descriptive graphs or as rigid, frozen constraints for dense artificial networks. **No prior work has performed biologically constrained structural remodeling across empirical connectomes to map the biological degeneracy frontier ($\Delta_{\mathbf{struct}}$ vs. $F_{\mathbf{task}}$), isolated non-rewirable "computational invariants" from developmental scaffolding, and translated these invariants into an asynchronous event-driven computing architecture (CoupledEventNet / CEN).**
4. **The Role of Human Data:** The human component serves as a **cross-validation and generalization tier**. H01 provides nanoscale micro-connectomic validation of synaptic motif invariants in human neocortex; HCP provides macro-scale topological comparison.
5. **Cross-Scale Neural Computational Motifs:** We formalize a 6-level validation hierarchy to test whether biological motifs represent genuine computational invariants that persist across scales.

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1. The Extended Biological Scale Ladder

```
flowchart TD
    subgraph ScaleLadder["The Five Biological Organizational Scales"]
        direction TB
        S1["<b>System A: Caenorhabditis elegans</b><br/>Scale: ~302 neurons, ~7,000
↪  synapses<br/>Scope: Complete whole-animal nervous system<br/>Data: EM anatomy,
↪  transmitters, developmental series (Cook 2019, Witvliet 2021)"]
        S2["<b>System B: Drosophila melanogaster</b><br/>Scale: 139,255 neurons, >50M
↪  synapses<br/>Scope: Complete adult central brain<br/>Data: EM wiring, 8,453 cell
↪  types, predicted transmitters, LIF model (FlyWire, Nature 2024)"]
        S3["<b>System C: Mus musculus (MICrONS)</b><br/>Scale: ~1,000 proofread neurons
↪  (1 mm3 volume)<br/>Scope: Primary visual cortex (V1 Layer 2/3) microcircuit<br/>Data:
↪  EM connectivity + in vivo 2-photon calcium imaging + visual stimulus (Nature 2025)"]
        S4["<b>System D (Micro): Homo sapiens (H01)</b><br/>Scale: ~16,000 neurons, ~150M
↪  synapses (1 mm3 volume)<br/>Scope: Anterior temporal cortex
↪  micro-connectome<br/>Data: Nanoscale EM reconstruction, 104 proofread cells, laminar
↪  E/I ratios (Science 2024)"]
        S5["<b>System D (Macro): Homo sapiens (HCP)</b><br/>Scale: 180 to 1015 cortical
↪  parcels<br/>Scope: Whole-brain macroscale network (1,200 subjects)<br/>Data:
↪  Diffusion tractography, resting/task fMRI, Connectome Workbench (NeuroImage 2012)"]
        S1 --> S2 --> S3 --> S4 --> S5
    end
```

```
    ScaleLadder --> Synthesis["Cross-Scale Invariant Discovery<br/>(6-Level Motif
↪   Verification)"]
    Synthesis --> RemodelingFrontier["Biologically Constrained Remodeling<br/>(The
↪   Degeneracy Frontier: Max Struct Dist vs Task Preservation)"]
    RemodelingFrontier --> CoupledEventNet["CoupledEventNet (CEN)<br/>Asynchronous
↪   Event-Driven Computing Architecture"]
```

Comprehensive Cross-Scale Dataset Comparison

Feature / Dimension	System A: *C. elegans*	System B: *Drosophila*
Organism / Strain	*C. elegans* (N2)	*D. melanogaster* (adult female)
Anatomical Scope	Complete whole-animal nervous system	Whole adult central brain
Node Count	302 (hermaphrodite) / 385 (male)	139,255 proofread neurons
Edge / Synapse Count	~7,000 chemical + ~900 gap junctions	54.5 million synapses
Spatial Resolution	Nanoscale EM (<5 nm)	Nanoscale EM (4x4x40 nm)
Primary Datasets	Cook et al. (2019); Witvliet (2021)	FlyWire v630 (2024)
Cell Types & Valences	Sensory, Inter, Command, Motor; GABA/Ach/Glu	8,453 cell types; Ach(+), Glu(-), GAB
Activity Data	Pan-neuronal calcium imaging (Kato 2015)	Functional calcium + optogenetic refle
Stimulus / Behavior	Chemotaxis, foraging, backward/forward	Antennal grooming, proboscis extensio
Simulation Infrastructure	OpenWorm, NeuroML, Python ODEs	Shiu LIF model (`Drosophila_brain_`
Access & License	Open / MIT (ConnectomeToolbox)	Open / CC BY 4.0 (Codex FlyWire)

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2. Deep Dive: Human Neural Data

A. The H01 Human Cortical Micro-Connectome

The H01 dataset (Shapson-Coe et al., *Science* 2024) represents the premier nanoscale reconstruction of human neocortex:

- **Tissue Origin & Region:** Resected anterior temporal lobe (gyrus temporalis medius/inferior) from a 45-year-old female patient undergoing surgery for deep hippocampal epilepsy. Neuropathological evaluation confirmed normal neocortical lamination and absence of cortical dysplasia.
- **Physical Volume & Sections:** Approximately 1 mm^3 ($3.0 \times 1.6 \times 0.2$ mm), cut into 5,300 serial sections at ~30 nm thickness on tape, imaged at 4 nm in-plane resolution using a multibeam scanning electron microscope (SEM).
- **Segmentation & Proofreading Quality:**
  - Automated segmentation via Google flood-filling networks (FFNs).
  - Agglomeration levels: C2 (aggressive agglomeration) and C3 (conservative agglomeration, minimizing false mergers; recommended for connectomic analysis).
  - 104 manually proofread neurons with complete axonal arbors, dendrites, and spine annotations.
  - Deep learning synapse detection identifying ~150M synapses classified into excitatory (asymmetric) vs. inhibitory (symmetric) with ~90% precision.
- **Novel Biological Discoveries in H01:**
  1. *Whorls of Axons:* Massive concentric spiral tangles of myelinated axons, unique to human cortex.
  2. *Ultra-Strong Multi-Synaptic Connections:* Individual axonal branches forming up to 50 distinct synaptic boutons onto a single target dendrite, far exceeding rodent microcircuit multi-synapse frequencies (typically 1–5 synapses).
  3. *Mirror-Image Pyramidal Pairs:* Pairs of pyramidal neurons whose apical dendrites point in polar opposite orientations.
  4. *Atypical Spiny Neurons:* Deep layer excitatory neurons displaying non-standard dendritic morphology.

- **Data Availability & Tooling:**
  - Publicly accessible via Google Cloud Storage (`gs://h01-release/`) under **Creative Commons Attribution 4.0 (CC BY 4.0)**.
  - Interactive browsing via Neuroglancer.
  - Programmatic access via TensorStore, CloudVolume, and Python (`ashapsoncoe/h01`).
  - Proofread subgraphs, edge lists, cell annotations (`agglo_20200916c3_cell_data.json`), and E/I ratio tables (`goog14r0s5c3_e_i_ratios_feb_2023.csv`) are cloned and verified locally in `upstream/human/h01/`.

B. The Human Connectome Project (HCP)

The Human Connectome Project (Van Essen et al., 2012; Glasser et al., 2016) is the gold-standard population-level human brain resource:

- **Scope:** 1,200 healthy young adult subjects scanned on customized 3T and 7T MRI scanners.
- **Data Modalities:** High-angular-resolution diffusion MRI (HARDI, 1.25 mm isotropic voxels), structural T1w/T2w imaging (0.7 mm), resting-state fMRI (2 mm, TR 720 ms), and task-evoked fMRI across 7 cognitive domains.
- **Parcellation & Matrix Representation:**
  - Multi-Modal Parcellation (Glasser MMP1.0): 180 cortical areas per hemisphere based on cortical architecture, function, connectivity, and topography.
  - Desikan-Killiany / Cammoun subdivisions: Multiscale parcellations ranging from 68 to 1,015 regions of interest (ROIs).
  - Deterministic and probabilistic tractography estimating the structural streamline density W_{ij} between all region pairs.
- **Why HCP is Fundamentally Different from H01:**
  - *Resolution:* HCP operates at the millimeter scale (10^{-3} m), where each voxel contains ~100,000 neurons and ~1 billion synapses. H01 operates at the nanometer scale (10^{-9} m), resolving individual synaptic vesicles and clefts.
  - *Topology:* HCP measures regional inter-area fiber tracts (macroscale routing). H01 measures synaptic connectivity between identified individual neurons (microcircuit computation).
  - *Sign / Directionality:* Standard diffusion tractography in HCP is **undirected** and cannot distinguish excitatory from inhibitory connections. H01 provides **directed** synaptic connections with explicit E/I classifications.

C. Additional Human Neural Resources

- **BICCN & BICAN (Brain Initiative Cell Census / Atlas Network):**
  - Hodge et al. (*Nature* 2019), Bakken et al. (*Nature* 2021): Single-nucleus RNA-seq revealing conserved cellular taxonomy between human and mouse cortex, but with significant evolutionary expansion of superficial layer (L2/3) pyramidal neurons.
  - Siletti et al. (*Science* 2023): Transcriptomic diversity of cell types across the adult human brain (3 million single-nucleus profiles).
- **Human Patch-seq Electrophysiology:**
  - Patch-clamp recordings in human neurosurgical slices (Allen Institute, Berg et al. 2021).
  - Provides empirical single-cell biophysical parameters: membrane time constants ($\tau_m \sim 15 - 30$ ms), input resistance, action potential width, and adaptation indices for human pyramidal and interneuron subtypes.

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3. Human Dataset Limitations & Forbidden Claims

To protect the scientific credibility of this research, we establish explicit **epistemic guardrails**:

Warning

Forbidden Over-Claim 1: Never state that "the human brain connectome is available."
Fact: A complete, neuron-by-neuron connectome of the human brain does NOT exist. We possess only macro-scale regional connectivity (HCP) and petavoxel fragments of single cortical areas (H01).

Warning

Forbidden Over-Claim 2: Never claim that "human cortex computes using the same algorithm as insects or nematodes."
Fact: Human cortex exhibits specialized evolutionary adaptations (e.g. 50-synapse connections, deep laminar expansion, diverse interneuron subtypes). We test for *conserved computational primitives*, not functional identity.

Warning

Forbidden Over-Claim 3: Never equate HCP macro-connectivity with synaptic micro-connectomics.
Fact: Voxel-level tractography does not measure synaptic wiring. Tractography streamlines represent macroscopic white-matter bundles, whereas micro-connectomics measures individual synaptic active zones.

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4. Definition: Cross-Scale Neural Computational Motif

We define a **Cross-Scale Neural Computational Motif** as a connectivity pattern that satisfies six progressively stringent, falsifiable criteria:

```
graph TD
    L1["Level 1: Structural Existence<br/>Sub-graph is empirically present in the
↪  connectome"] --> L2["Level 2: Statistical Significance<br/>Z > 2.0 vs
↪  degree-preserving null models"]
    L2 --> L3["Level 3: Dynamic Computation<br/>Measurably enhances task performance in
↪  event-driven simulation"]
    L3 --> L4["Level 4: Causal Invariance<br/>Lesioning drops task performance by >50%
↪  (non-rewirable bottleneck)"]
    L4 --> L5["Level 5: Architectural Transfer<br/>Useful in artificial neuromorphic
↪  network outside biological context"]
    L5 --> L6["Level 6: Cross-Scale Conservation<br/>Conserved across Worm, Fly, Mouse,
↪  and Human micro-connectomes"]
```

1. **Level 1 (Structural Existence):** The motif is identified within the directed adjacency matrix of the organism.
2. **Level 2 (Statistical Overrepresentation):** The motif occurs with a frequency significantly exceeding degree-preserving configuration null models ($Z > 2.0$, $p < 0.01$).
3. **Level 3 (Computational Contribution):** Under Leaky Integrate-and-Fire (LIF) dynamics, the motif executes a specific temporal operation (e.g., coincidence detection, duration discrimination, burst normalization) with higher fidelity than randomized graphs.
4. **Level 4 (Causal Invariance / Non-Rewirability):** Targeted lesioning of the motif's core edges causes a catastrophic collapse in computation ($> 50\%$ error increase), whereas lesioning peripheral scaffolding produces negligible impact ($< 2\%$).
5. **Level 5 (Architectural Transferability):** An abstraction of the motif confers measurable advantages (lower SynOps, higher latency efficiency) when instantiated in an artificial event-driven neural network.
6. **Level 6 (Cross-Scale Conservation):** The motif structure and its dynamical computational role recur across phylogenetic levels (e.g., from *C. elegans* locomotion to fly motion detection, mouse V1 orientation, and human H01 micro-connectivity).

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5. Upstream Repositories Inventory (Updated)

All authoritative upstream repositories have been cloned and cataloged with metadata descriptors in `upstream/`:

```
upstream/
├── openworm/
│   ├── ConnectomeToolbox/          (Commit: b9c0b4a7... | MIT License | C. elegans datasets)
│   └── CElegansNeuroML/            (Commit: b36380a3... | LGPL-3.0 | Biophysical cell models)
├── flywire/
│   ├── FlyConnectome/              (Commit: 9b1790b5... | GPL-3.0 | CAVE tooling)
│   ├── flywire_annotations/        (Commit: 8587524c... | GPL-3.0 | 139k neuron annotations)
│   ├── flywire-network-analysis/   (Commit: ee4944c0... | MIT License | Lin et al. Nature 2024)
│   └── Drosophila_brain_model/     (Commit: 91bdd1e7... | MIT License | Shiu et al. Nature 2024 LIF)
├── mouse/
│   ├── MicronsBinder/              (Commit: 3f53e35e... | Allen License | Pinky100 & mm^3 V1)
│   └── CAVEclient/                 (Commit: c57c15f5... | BSD-3-Clause | Live CAVE queries)
├── human/
│   ├── h01/                        (Commit: db223b8d... | CC BY 4.0 | Shapson-Coe Science 2024)
│   └── HCPpipelines/               (In progress | HCP License | Van Essen / Glasser 2016)
└── other/
    ├── conn2res/                   (Commit: 03046778... | BSD-3-Clause | Connectome reservoirs)
    └── ncps/                       (Commit: d90cf1f5... | Apache-2.0 | Neural Circuit Policies)
```

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6. Updated Prior Art & Novelty Matrix Summary

The comprehensive 18-column matrix in `research/prior_art_matrix.csv` has been expanded to include H01, HCP, and BICCN. Key prior-art classifications:

Category	Key Prior Art	Status	Our Contr
Static Cross-Species Graph Motifs	Lin et al. (*Nature* 2024); Milo et al. (2002)	**RED**	Lin et al.
Drosophila Whole-Brain LIF Simulation	Shiu et al. (*Nature* 2024)	**RED**	Shiu et al.
C. elegans Neural Circuit Policies	Hasani & Lechner (*Nat Mach Intell* 2020)	**RED**	Hasani et
Connectome-Informed Reservoirs	Suárez, Misic et al. (*Nat Commun* 2024)	**RED**	`conn2res`
Human H01 Micro-Connectomics	Shapson-Coe et al. (*Science* 2024)	**RED**	Shapson-C
Event-Driven Cross-Scale Architecture	*None found*	**GREEN**	**Genuinel**
Biologically Constrained Remodeling Frontier	*None found*	**GREEN**	**Genuinel**

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7. Proposed Candidate Experiments (Incorporating Human Data)

Candidate Experiment 1: The Cross-Scale Biological Remodeling Frontier (Δ_{struct} vs. F_{task})

- **Goal:** Measure how far biological neural subcircuits can be structurally deformed under Dale's principle while preserving dynamic task performance.
- **Scope:**
  - *C. elegans:* Chemosensory-command circuit (25 neurons, Cook 2019).
  - *Drosophila:* Optic lobe motion circuit (60 neurons, FlyWire v630).
  - *Mouse:* Layer 2/3 orientation microcircuit (50 neurons, MICrONS).
  - *Human:* Layer 2/3 temporal cortex microcircuit (50 neurons, H01 proofread cells).
- **Method:** Multi-objective evolutionary search (NSGA-II) optimizing:

$$\max \Delta_{\text{struct}}(G_{\text{orig}}, G'), \quad \min \text{SynOps}(G') \quad \text{subject to} \quad F_{\text{task}}(G') \geq (1-\epsilon) F_{\text{task}}(G_{\text{orig}})$$

enforcing: Dale's principle (no E/I sign flipping), bounded delays ($1 - 20$ ms), and degree caps.
- **Outcome:** Quantify the maximum permissible synaptic turnover (Δ^*_{struct}) across species, identifying the non-rewirable invariant core.

Candidate Experiment 2: Causal Verification of Cross-Scale Invariant Motifs (Level 1–6)

- **Goal:** Rigorously validate candidate motifs (cFFL, iFFL, Reciprocal Inhibitory Pair, Disinhibitory Triad) across all four micro-connectomes (*C. elegans*, fly, mouse, human H01).
- **Method:**
  1. Count motif frequencies and Z-scores against degree-preserving configuration nulls (Levels 1 & 2).
  2. Embed isolated motifs in a LIF testbed to perform temporal coincidence detection, duration discrimination, and gain normalization (Level 3).
  3. Perform causal edge lesioning to evaluate performance collapse (Level 4).
  4. Compare against 4 strict null models (Erdős-Rényi, configuration null, shuffled-weight, shuffled-sign).
- **Outcome:** Identify which motifs achieve full Level 1–6 validation as **Cross-Scale Neural Computational Motifs**.

Candidate Experiment 3: CoupledEventNet (CEN) Architecture Benchmarking

- **Goal:** Synthesize the validated cross-scale invariants into an asynchronous event-driven architecture and benchmark against state-of-the-art baselines.
- **Baselines:**
  1. Dense Leaky Integrate-and-Fire SNN (surrogate gradient BPTT).
  2. Connectome Reservoir (`conn2res` using C. elegans, Drosophila, and HCP human consensus connectomes).
  3. Neural Circuit Policies (NCPs / LTCs from Hasani et al.).
  4. Standard Recurrent Neural Network (LSTM / GRU).
- **Benchmark Tasks:** Sequential XOR, Delay-Match-to-Sample, Spatio-Temporal Spike Pattern Classification (neuromorphic event streams).
- **Metrics:** Accuracy, F1-score, total Synaptic Operations (SynOps), latency to decision, event sparsity, parameter count, and noise robustness.

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8. Recommended Implementation Roadmap

We recommend an integrated **three-phase experimental progression**:

Phase 1 (Causal Motif Discovery & H01 Validation) ⟶ Phase 2 (Biologically Constrained Remodeling Frontier) ⟶ Phase

Immediate Next Implementation Actions:

1. **Connectome Data Adapters (`src/connectome/`):** Implement loaders for *C. elegans* (Cook 2019), *Drosophila* (FlyWire v630), *Mus musculus* (MICrONS V1), *Homo sapiens* H01 (proofread cells), and *Homo sapiens* HCP (consensus connectivity).
2. **Unified Spiking Simulation Engine (`src/simulation/`):** Vectorized asynchronous Leaky Integrate-and-Fire (LIF) simulator supporting biophysical delays, Dale's principle, and SynOp tracking.
3. **Execution of Phase 1:** Run motif extraction across all four micro-connectomes and evaluate dynamical performance against configuration nulls.